

Proposal: Independent Temperature-Controlled Fan System

Local thermal stabilisation around the camera / heat-sink assembly

1. Aim

I propose adding a small independent fan-control system around the camera / heat-sink region to reduce local thermal excursions without running the fans at maximum speed continuously. The controller would be separate from the LK220 and from the camera TEC. Its only purpose would be to respond to the local surface temperature and provide additional airflow when required.

2. Proposed operating principle

The first version would use a simple two-level control strategy rather than a full PID loop. Under normal thermal conditions the two 5 V fans would operate at approximately 50% command. If the measured local temperature rises above a defined high-temperature threshold, both fans would switch to 100%. Once the temperature falls below a lower return threshold, the fans would return to the normal reduced-speed state.

This creates a small hysteresis band so that the controller does not rapidly switch between the two states. The final temperature thresholds would be selected from measured operating data rather than assuming that the correct camera-body temperature is exactly the LK220 coolant setpoint.

3. Why I am proposing an independent controller

The camera CMOS TEC and the LK220 already have their own control loops. I therefore want the fan system to remain independent and deliberately simple. The fan controller would not alter the LK220 setpoint, PID parameters, pump or fan settings. It would only monitor a local surface-temperature sensor and adjust airflow.

The PT104, BME680 and LK220 telemetry can continue to be logged during testing so that the effect of the fan system can be quantified independently.

4. System architecture

A small local temperature sensor would be attached to the camera / heat-sink region. An Arduino Nano would read this sensor and command a logic-level MOSFET that supplies both fans. The fans would be powered from a separate regulated 5 V supply, with the Arduino and sensor using the same low-voltage supply.

Independent fan controller - veroboard layout (component side)

Horizontal copper strips are on the underside. One track cut is required on row A at column 4 for the fuse.

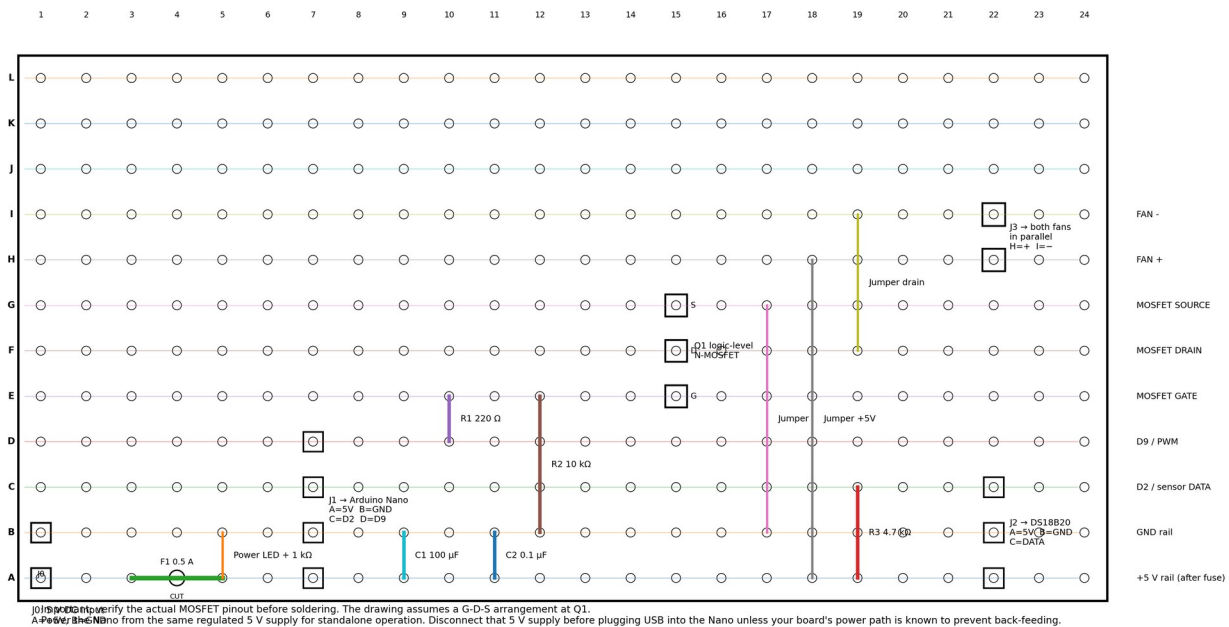


Figure 1. Proposed Arduino / MOSFET / veroboard implementation.

5. Proposed components

Qty	Component	Specification	Purpose
2	Fans	SUNON MF30100V3-1000U-A99, 5 V	Local forced-air cooling
1	Microcontroller	Arduino Nano / compatible	Temperature logic and fan control
1	Temperature sensor	DS18B20 probe	Independent surface-temperature measurement
1	MOSFET	Logic-level N-channel	Switches both fans
1	Power supply	Regulated 5 V, ≥ 1 A	Powers controller and fans
1	Fuse / polyfuse	~ 0.5 A	Input protection
1	Resistor	220 ohm	PWM pin to MOSFET gate
1	Resistor	10 kohm	MOSFET gate pull-down
1	Resistor	4.7 kohm	Temperature-sensor data pull-up
1	Capacitor	100 uF, ≥ 10 V	Fan-rail bulk decoupling
1	Capacitor	0.1 uF ceramic	High-frequency decoupling

1	Veroboard / stripboard	Approx. 24 x 12 holes or larger	Permanent prototype assembly
1 set	Jumper wires	Male-male / male-female / solid-core	Interconnections
2-3	Connectors	2-pin / 3-pin terminal or JST	Removable fan/sensor/power wiring
1	USB cable	Arduino compatible	Programming / test logging

A separate MOSFET heatsink should not be required for this load because the two fans together draw only a small current. The existing camera / heat-sink structure remains the thermal element being cooled.

6. Control logic

The initial control logic would be:

- Startup: 100% fan command for approximately 1-2 s to ensure both fans start reliably.
- Normal state: approximately 50% fan command.
- Hot state: 100% fan command when the measured surface temperature reaches T_{hot} .
- Return state: back to 50% only when the temperature falls below T_{return} .
- Sensor fault: default to 100% fan command as a fail-safe.

7. Experimental validation

Before using automatic control, I would first characterise the thermal effect of the fans at fixed operating points. The simplest comparison would be fan OFF, reduced-speed operation and full-speed operation while keeping the LK220 and camera settings unchanged.

For each condition I would compare the calibrated PT104 temperatures, local temperature gradients, peak-to-peak excursions and the known room/AC periodic component. The purpose is to determine whether controlled airflow reduces local thermal variation and whether the reduced-speed state is sufficient during normal operation.

8. Veroboard implementation

The proposed stripboard layout uses horizontal copper strips on the underside. The protected +5 V rail supplies the Arduino, temperature sensor and both fans. The Arduino D9 output drives the MOSFET gate through a 220 ohm resistor, with a 10 kohm pull-down to ground. The DS18B20 data line is connected to D2 with a 4.7 kohm pull-up to +5 V. Both fans are connected in parallel and switched on the low side by the MOSFET.

Before soldering, the exact MOSFET package pinout must be checked because Gate-Drain-Source order differs between devices. All grounds must be common.

9. Expected advantages

The main advantage of this approach is that it introduces local airflow only when needed. It avoids running the fans at maximum speed continuously, which should reduce unnecessary vibration, acoustic noise and power consumption. It also keeps the additional control system separate from the LK220, making it easier to test whether airflow itself improves the thermal behaviour.

10. Decision criteria

I would consider the concept successful if the fan system measurably reduces camera-body temperature excursions and thermal gradients, while remaining mechanically stable and not introducing a new measurable disturbance. If the reduced-speed state is not reliable with the selected two-wire fans, the normal duty can be increased or a dedicated PWM-control fan variant can be used without changing the overall controller architecture.

11. Proposed next step

Build the controller first on a breadboard or temporary wiring arrangement, verify the temperature sensor and both fan states away from the instrument, and then install it temporarily for thermal testing. Once the thermal benefit and appropriate fan duty are established, transfer the circuit to veroboard and finalise the mechanical mounting.